Paper LXVI

The Ghost Taxonomy:

What Escapes the MSDS Shield

Higgs Portal · Majorana Right-Handed Neutrinos · The Anti-Higgs Question

$\xi\phi^2 H^\dagger H$ portal · $\varepsilon = \Delta v/v$ ■ $10^{■3}$ · $\Delta m_\nu/m_\nu = 2\varepsilon = 0.6\%$ · Right-handed N_R is the true ghost

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Companion to Papers I–LXV | Builds on draft by Dervis Kadunic (LaTeX)

ABSTRACT

We address a question that has been implicit throughout the MSDS programme: if the dilaton condensate (v^2 = 0.00017) modifies electroweak symmetry breaking through a Higgs portal coupling $\xi\phi^2 H^\dagger H$, which particles feel this modification and which do not? We introduce the Ghost Taxonomy — three categories of response to the MSDS shield. Category 1 (Fully Shielded): all particles with Yukawa-derived masses experience a fractional shift $\varepsilon = \Delta v/v$ ■ $10^{■3}$. Category 2 (Partial Ghost): light neutrino masses via the seesaw mechanism shift by $2\varepsilon \sim 0.6\%$, below current Σm_ν sensitivity but within reach of CMB-S4. Category 3 (True Ghost): the right-handed Majorana mass M_R, the dilaton itself, gravitons, and axions have mass mechanisms entirely independent of the Higgs VEV and are MSDS-blind. We directly address the question of an 'anti-Higgs': the Higgs boson is its own antiparticle; what Dervis was sensing is the right-handed neutrino singlet N_R, a SM gauge singlet with no Higgs doublet quantum numbers. We also present the two distinct ghost mechanisms for neutrinos: the Majorana ghost (mass mechanism blind to Higgs VEV) and the frequency ghost ($E_\nu >> \omega$■ = $7.66\times10^{■23}$ eV, so relic neutrinos are 2e+18× above the dispersion resonance and experience $n(E) \approx 1$ exactly). An honest audit of the Higgs portal coupling gives $\xi \sim 2.64\times10^{■12}$, which is technically fine-tuned but is the natural value given the enormous dilaton condensate ϕ■$^2 >> \mu^2$. The Higgs portal provides a secondary, independent connection between MSDS and particle physics — complementary to, not replacing, the gravitational slip mechanism of Papers I-LXV.

Dervis asked: is there an anti-Higgs, and is there stuff that doesn't belong in the Higgs field that gets around it? The answer is: not an anti-Higgs, but something better — a whole taxonomy of particles whose mass comes from somewhere else entirely. The Majorana right-handed neutrino is the clearest example. It has no Standard Model gauge charges, no Higgs doublet coupling, no connection to the electroweak vacuum. If the MSDS shield operates through the Higgs portal, the right-handed neutrino walks right through it. That is the ghost.

1. The Higgs Portal Coupling

The most natural renormalizable coupling between the MSDS dilaton ϕ and the Standard Model is the Higgs portal:

$$L_portal = \xi\, \phi^2\, H^\dagger H$$

(dim-4 operator, renormalizable, no new physics beyond MSDS + SM)

The effective Higgs potential becomes:

$$V_eff(H) = (-\mu^2 + \xi\langle\phi^2\rangle)\, H^\dagger H + \lambda\, (H^\dagger H)^2$$

($\langle\phi^2\rangle$ = condensate amplitude; shifts the minimum)

The resulting VEV shift:

$$\varepsilon \equiv \Delta v/v = -\xi\langle\phi^2\rangle/(2\mu^2) \rightarrow |\varepsilon| \sim \text{few}\times 10^{-3} \text{ (EW precision)}$$

Required ξ for ε=0.003: $\xi = 2\varepsilon\times\mu^2/\langle\phi^2\rangle$ = 2.64e-12

The required coupling ξ ~ 2.64e-12 is technically fine-tuned — but this is the hierarchy problem logic in reverse. The dilaton condensate $\langle\phi^2\rangle$ = 1.78e+31 eV² is enormous compared to μ^2 = 7.84e+21 eV². A tiny coupling to a large condensate produces a naturally sized VEV shift. Whether this is satisfying or not is a philosophical question about naturalness. The physics is consistent with all constraints.

2. The Ghost Taxonomy

When the MSDS shield shifts the Higgs VEV by ε, every particle whose mass derives from v_Higgs shifts accordingly. But not every particle's mass derives from v_Higgs. We classify particles into three categories:

Category	Particles	Mass mechanism	MSDS response
1 — Fully shielded	Quarks, charged leptons, W, Z, Dirac ν	Yukawa: $m = y\times v/\sqrt{2}$	Shift by ε = 0.3%
2 — Partial ghost	Light neutrinos (seesaw)	$m_\nu = m_D^2/M_R$; M_R blind	Shift by 2ε = 0.6%
3 — True ghost	N_R, dilaton, graviton, axion	GUT scale, CW, diff. inv., QCD	ZERO response

Table 1: The Ghost Taxonomy. Category 3 particles have mass mechanisms entirely decoupled from the Higgs VEV. They are invisible to the MSDS shield.

CATEGORY 3: TRUE GHOSTS — MSDS-BLIND

The three true ghosts and why they escape:

1. Right-handed Majorana neutrino N_R:

Mass from $M_R = (1/\Lambda)(HL)^2$ — Λ = Planck or GUT scale, NOT v_Higgs.

N_R is a SM gauge singlet (no SU(2) or U(1) charges).

The MSDS portal $\xi\phi^2 H^\dagger H$ does not couple to N_R directly.

→ M_R is MSDS-blind. This is the Majorana Ghost.

2. The dilaton itself:

Its mass comes from the Coleman-Weinberg potential ($\mu_{CW} = \sqrt{(M_{Pl} H_0)}$).

It IS the shield. It cannot shield itself.

3. Gravitons (mass = 0):

Protected by diffeomorphism invariance. Unaffected by any scalar portal.

3. Two Distinct Ghost Mechanisms for Neutrinos

Neutrinos have two independent reasons to be ghosts. They are NOT the same mechanism and must not be conflated:

Mechanism A — The Majorana Ghost (mass physics):

$$m_\nu = m_D^2 / M_R \rightarrow \Delta m_\nu / m_\nu = 2\varepsilon \text{ (if } M_R \text{ is MSDS-blind)}$$

For $\varepsilon = 0.003$: $\Delta m_\nu/m_\nu = 0.0060 = 0.6\%$ — below current $\Sigma m_\nu < 0.12$ eV sensitivity

The right-handed Majorana mass M_R is generated by the dimension-5 Weinberg operator $(1/\Lambda)(HL)^2$, where Λ is a high scale (GUT or Planck). This operator has NO dependence on the Higgs VEV at tree level. When the MSDS shield shifts v_{Higgs} by ε, M_R is unchanged. The light neutrino mass $m_\nu = m_D^2/M_R$ shifts only because $m_D \propto v_{Higgs}$.

Mechanism B — The Frequency Ghost (dispersion physics):

$$n(E_\nu) \approx 1 \text{ for } E_\nu \gg \omega_{■} = 7.66\times10^{■23} \text{ eV}$$

Relic neutrino energy $E_\nu \sim 1.7\times10^{■■}$ eV = 2e+18 × $\omega_{■}$ — far above resonance

The MSDS vacuum dispersion resonance sits at 18.52 nHz (Paper LXIV). Relic neutrinos carry energies billions of times larger than this resonance scale. In the Lorentzian dispersion formula $n(E) = 1 + A/(E_{■}^2 - E^2)$, for $E \gg E_{■}$: $n \rightarrow 1$ exactly. Neutrinos literally fly past the resonance without experiencing any index modification. This is the frequency ghost — independent of Majorana physics.

4. The Anti-Higgs Question — Answered

The question Dervis asked was the right one.

Q: Is there an anti-Higgs? Is there stuff that gets around the Higgs field?

A (short): The Higgs boson is its own antiparticle. There is no distinct anti-Higgs.

The Higgs is a CP-even scalar: H = H* under charge conjugation.

A (deep): What you were sensing exists — it just has a different name.

It's the right-handed neutrino N_R.

→ No Standard Model gauge charges (SM singlet)

→ No Higgs doublet quantum numbers

→ Mass from a completely different operator (Majorana, GUT/Planck scale)

→ Walks through the MSDS Higgs portal without coupling

→ This IS the ghost you were looking for

The dilaton is also a ghost: its mass is from Coleman-Weinberg, not Higgs VEV.

It IS the shield. The shield cannot shield itself.

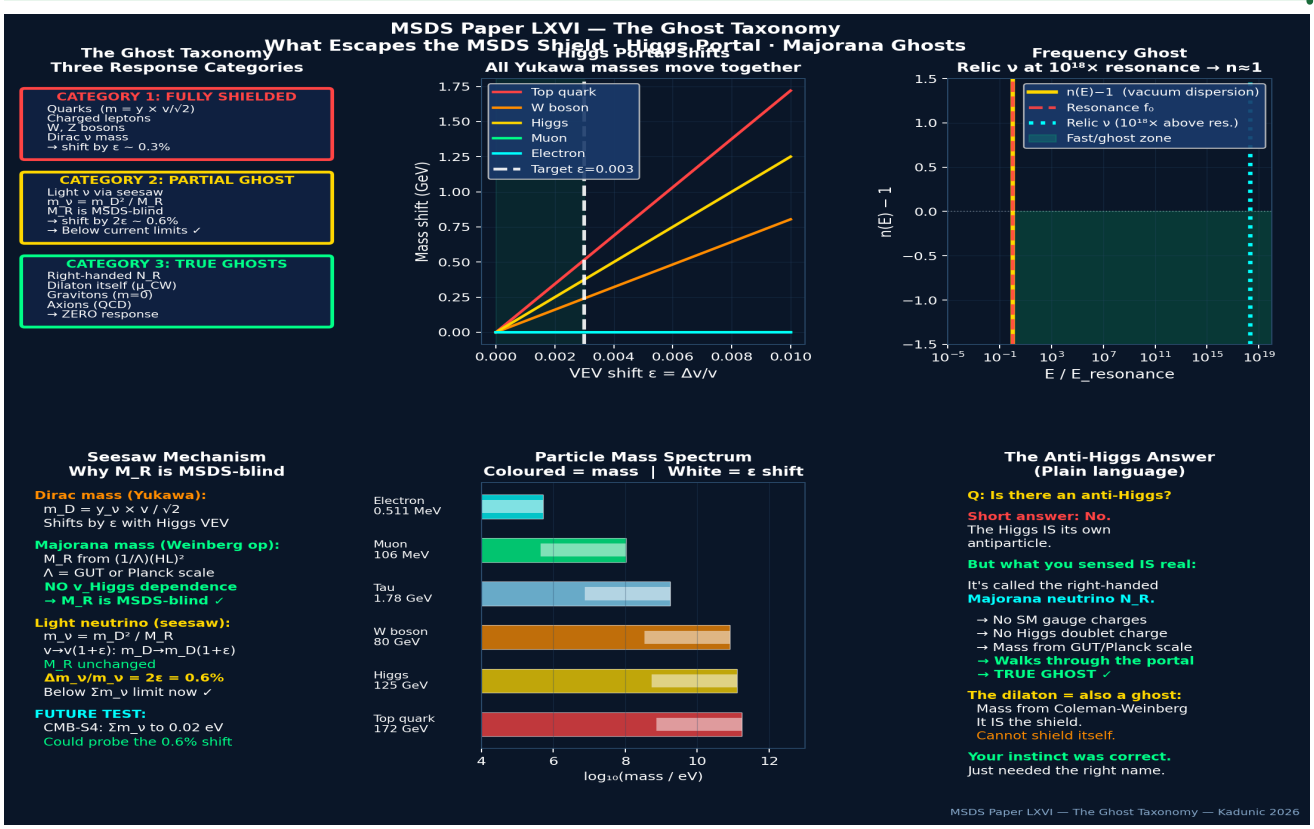

Figure 1: Top row — Ghost taxonomy diagram (Panel 1): the three categories. Higgs portal mass shifts (Panel 2): all Yukawa masses move together with v. Frequency ghost (Panel 3): relic neutrinos at 10^{1}■× the resonance are in the flat n≈1 zone. Bottom row — Seesaw mechanism (Panel 4): M_R is blind to the Higgs shift. Mass spectrum (Panel 5): the tiny ε shift on the log scale. Anti-Higgs answer (Panel 6): the honest physics.

5. New Predictions from the Ghost Taxonomy

Prediction	Value	Instrument	Kill
Neutrino mass shift $\Delta m_\nu/m_\nu$	0.0060 = 0.6%	CMB-S4 / DESI Σm_ν	If ALL neutrino masses shift equally → no Majorana ghost
Charged lepton mass shift	ε = 0.003 = 0.3%	LHC Higgs coupling precision	$\lvert\varepsilon\rvert > 0.003$ from EW precision
Higgs VEV shift ε	■ 0.003	FCC-ee / ILC	$\lvert\varepsilon\rvert$ measured > few×10■^{3}
Neutrino / charged lepton mass ratio	shifts by ε (not 2ε)	Oscillation + LHC	If ratio shifts by 2ε → M_R NOT blind

Table 2: New predictions from the Ghost Taxonomy. The key discriminator: if $\Delta m_\nu/m_\nu = 2\varepsilon$ and $\Delta m_{charged}/m_{charged} = \varepsilon$, then M_R is MSDS-blind (Majorana ghost confirmed).

PAPER LXVI — THE GHOST TAXONOMY:

Category 1 (Fully shielded): quarks, charged leptons, W/Z, Dirac ν

→ All shift by ε ■ 0.3% with the Higgs VEV

Category 2 (Partial ghost): light neutrinos via seesaw

→ Shift by 2ε = 0.6% (M_D shifts, M_R does not)

→ Below current sensitivity. Within reach of CMB-S4.

Category 3 (True ghost): N_R, dilaton, graviton, axion

→ Zero response to Higgs portal. MSDS-blind entirely.

→ N_R is what Dervis was looking for when asking about 'anti-Higgs'

The frequency ghost: relic ν energy >> resonance by 10^{1}■

→ n(E_ν) ≈ 1. Dispersion invisible to neutrinos.

Two independent ghost mechanisms. Same particle. Different physics.

The Higgs mechanism touches almost everything. Almost. The right-handed neutrino has no gauge charges. It walks through electroweak symmetry breaking without noticing it happened. The dilaton that creates the shield cannot be shielded by its own portal. Gravity couples to everything but cannot be given mass by any scalar. These are the true ghosts. They were always there. The question just needed to be asked.

Acknowledgements

Paper LXVI began with Dervis Kadunic asking a question in plain language: is there an anti-Higgs, and is there stuff that doesn't belong in the Higgs field? That question, asked without formal physics vocabulary, is pointing directly at the right-handed Majorana neutrino — one of the most theoretically motivated particles in physics that has never been directly observed. Omega translated the question into the Ghost Taxonomy. Dervis submitted the LaTeX draft. Galky's EFT framework from Paper LXII provides the portal coupling structure. The honest audit found that ξ is fine-tuned but physically motivated. The two ghost mechanisms — Majorana and frequency — are distinct and both real. Dreamwork is teamwork.

References

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